

Assignment 2

Real world Application: online video streaming platform
(e.g., youtube, Netflix)

1) use of parallel computing. parallel computing is used in video encoding, recommendation system, & real time analysis.

- Video Encoding: videos are split in chunks & encoded in parallel using GPUs multi-core CPUs. This speeds up processing for large volumes of uploaded content.
- Recommendation: Its models for suggesting content use parallel processing to handle massive datasets quickly.
- Analytics :- Real time viewership data is processed in parallel to monitor performance.

2) use of Networked Systems: Networked system delivers video content from distributed servers (CDNs - content delivery Networks) to users worldwide.

- Client Server architecture: clients (users) request video, ~~server~~ deliver based on location.
- Load balancing: traffic is distributed among servers to prevent overload.
- Reliability: if one server fails, others handle to request seamlessly.

Why its important:

- Ensures global availability & fast deliveries
- Enhances user experience with minimal buffering
- Supports millions of concurrent users efficiently.